

⚡ PHYSICS CH.8 — WORK, ENERGY AND POWER — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

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1. Electric Power: Formulas, Units & Domestic Supply

- Electric power ke valid formulas: $P = VI = I^2R = V^2/R$; $P = IR^2$ galat hai. [Q1210]
- Power ki definition: rate of doing work ya rate of energy transfer hai; power = Work/Time hai. [Q1234]
- Electric power bhi isi definition ka special case hai — rate at which electrical energy dissipate/consume hoti hai circuit mein. [Q1217]
- Resistance ka formula work se: $R = W/(I \times q)$ kyunki $W = I^2Rt = IRq$. [Q1216]
- Charge Q, voltage V, time t se power: $P = VQ/t$ kyunki $I = Q/t$. [Q1215]
- Circuit mein heat loss: $H = I^2RT$ — ye directly proportional to resistance hai same current ke liye. [Q1214]
- Electrical energy consumed: Power rating $\times$ Time par depend karta hai. [Q1224]
- Bulb ki power rating 100 W ka matlab: bulb 100 Joule energy per second consume karta hai (na ki per minute ya per hour). [Q1226]
- 1 joule/second = 1 watt — power ka SI unit. [Q1268]
- Domestic supply India mein: 220V, 50 Hz; USA mein 110V, 60 Hz. [Q1213]
- 1 kW = 1000 W; 1 kWh = 3.6×10^6 J (commercial unit of electrical energy). [Q1246, Q1252, Q1311]
- Household/industry/commercial energy consumption usually kilowatt-hour (kWh) mein express hoti hai. [Q1272]
- 1 HP (Horsepower) = 746 W. [Q1311]

2. Energy Conversions & Devices

- Electric motor: Electrical energy → Mechanical energy. [Q1211]
- Dynamo/Generator: Mechanical/Kinetic energy → Electrical energy. [Q1222]
- Electric geyser: Electrical energy → Heat energy (Joule heating). [Q1212]
- Electric lamp: Electrical energy → Heat + Light energy. [Q1211]
- Thermocouple: Heat energy → Electrical energy. [Q1211]
- Microphone: Sound energy → Electrical energy. [Q1211, Q1310]
- Photocell: Light energy → Electrical energy. [Q1222]
- Battery/Cell: Chemical energy → Electrical energy. [Q1299, Q1310]
- Electric heater: Electrical → Heat; Hair dryer: Electrical → Thermal; Electric fan: Electrical → Mechanical. [Q1299]
- Thunder: Atmosphere ke charged particles ki KE → Light (lightning) + Sound energy mein convert hoti hai. [Q1286]

3. Energy Sources: Renewable, Non-Renewable & Conventional

- Thermal power plants mein fossil fuel ki Chemical energy → Electrical energy mein convert hoti hai. [Q1229]
- Hydroelectric power mein falling water ki PE → KE → Mechanical → Electrical hoti hai. [Q1231, Q1245, Q1285, Q1307]
- Wind turbines wind ki KE → Mechanical → Electrical mein convert karte hain. [Q1219, Q1230]
- India ka sabse bada wind farm: Muppandal (Kanyakumari), 1500 MW; World ka sabse bada: Gansu Wind Farm, China. [Q1219]
- Solar energy collectors mein sabse zyada efficiency paraboloid dish collector ki hoti hai kyunki ye sunlight ko focal point par concentrate karta hai. [Q1287]
- Calorific value: 1 kg fuel ke complete combustion se kitni heat milti hai; unit kJ/kg. [Q1227]
- Conventional (Non-renewable): Natural gas, coal, petroleum, firewood — limited hain aur pollution karte hain. [Q1281]
- Non-conventional (Renewable): Solar, wind, nuclear, hydroelectric, geothermal, tidal, bioenergy. [Q1281, Q1230]

4. Kinetic Energy: Basics, Formula & Factors

- Kinetic energy formula: $KE = \frac{1}{2}mv^2$ — mass aur velocity ke square par depend karti hai. [Q1228, Q1232, Q1236, Q1237, Q1239, Q1242, Q1244, Q1257, Q1261, Q1274, Q1277, Q1279, Q1297, Q1301, Q1313]
- KE hamesha zero ya positive hoti hai; ye scalar quantity hai. [Q1239, Q1279]
- Velocity double karne par KE 4 times; 4 times karne par 16 times ho jaati hai. [Q1257]
- Same momentum par: $KE = \frac{p^2}{2m}$ — jitna kam mass utni zyada KE; isliye bullet ki KE gun se zyada hoti hai. [Q1242, Q1266]
- Free fall mein KE continuously badhti hai kyunki velocity badhti hai. [Q1237, Q1244]
- Moving bullet, railway engine, simple pendulum, running water, fast asteroid, car, ocean waves — sab mein KE hoti hai. [Q1239, Q1254, Q1277, Q1285]
- Bullet laksya mein gahraai tak penetrate karti hai kyunki usme bohot zyada KE hoti hai. [Q1261]
- Moving bullet sabse zyada kaam kar sakti hai compared to raised hammer, speeding stone, ya rotating wheel. [Q1313]
- Chalti car ka velocity badhne par uski PE stable rehti hai kyunki height nahi badalti. [Q1293]
- Gas molecules ki KE directly proportional to temperature hoti hai. [Q1223]
- States of matter mein particles ki KE order: Solids < Liquids < Gases < Plasma. [Q1238, Q1290]

5. Potential Energy, Mechanical Energy & Conversions

- Potential energy formula: $PE = mgh$ — position, height ya shape par depend karti hai. [Q1232, Q1241, Q1250, Q1260, Q1263, Q1270, Q1283, Q1302, Q1305]
- Kisi bhi object ki energy (uski work karne ki ability), uske position aur condition par depend karti hai — na ki sirf mass/volume ya direction par. [Q1267]
- Gravitational PE ke examples: ghar ki chhat par rahi eent, dam ke upar stored paani, uthaya hua hathaura. [Q1283, Q1245, Q1312]
- Elastic PE ke examples: stretched rubber band, compressed spring, wound spring, compressed slinky, stretched bow. [Q1259, Q1243, Q1283, Q1294, Q1302, Q1305, Q1309]
- Object ko vertically up throw karne par: jaise-jaise height badhti hai PE increases, KE decreases. [Q1241]
- Maximum height par $KE = 0$ aur $PE = \text{maximum}$ hoti hai. [Q1250]
- $3/4$ height par bhi $PE > KE$ rehti hai. [Q1250]
- 200 m se girte object ko 100 m girne ke baad aadhi PE aur aadhi KE hoti hai (donon barabar). [Q1280]
- Mechanical energy = $KE + PE$; free fall mein (air resistance ignore karke) constant rehti hai. [Q1244, Q1249, Q1265, Q1280, Q1298]
- Raindrops neeche girte waqt $PE \rightarrow KE$ mein convert hoti hai. [Q1275]
- Arrow release hone par bowstring ki $PE \rightarrow$ arrow ki KE mein convert hoti hai. [Q1247, Q1309]
- Compressed slinky release karne par $PE \rightarrow KE$ hoti hai. [Q1294]
- Bullet fire hone par stored $PE \rightarrow KE$ mein convert hoti hai. [Q1308]
- Pendulum ka example: highest point par max PE, min KE; lowest point par max KE, min PE — continuous $PE \leftrightarrow KE$ conversion hota hai. [Q1314]

6. Work Done: Definition, Formula, Conditions & Types

- Work done formula: $W = F \cdot d = F \times d \times \cos\theta$; unit joule (J) = newton $\times$ meter; dimension $[ML^2T^{-2}]$. [Q1248, Q1255, Q1311]
- 1 joule ki definition: 1 newton $\times$ 1 meter (na ki watt $\times$ meter ya pascal $\times$ meter). [Q1258]
- Work karne ke liye do shartein: (1) force lagna chahiye, (2) displacement hona chahiye. [Q1262, Q1276, Q1291]
- Work done positive jab force aur displacement same direction mein ho $\rightarrow$ body energy gain karti hai. [Q1240, Q1251]
- Work done negative jab force aur displacement opposite direction mein ho. [Q1251, Q1271]
- Work done zero jab force aur displacement perpendicular ho ($\theta = 90^\circ$, $\cos 90^\circ = 0$) ya koi displacement hi na ho. [Q1235, Q1273, Q1278, Q1284, Q1296, Q1303, Q1304, Q1306]
- Zero work ke examples: wall push karna (displacement nahi), platform par khade rehna, circular motion (centripetal force perpendicular), Moon ki Earth parikrama (gravitational force perpendicular), gadhe dwara bojh lekar chalna (vertical force, horizontal displacement). [Q1278, Q1284, Q1296, Q1303, Q1304, Q1306]
- Friction ka work hamesha negative hota hai kyunki ye hamesha motion ka virodh karta hai. [Q1289]
- Work done mass par depend nahi karta; sirf force, displacement, aur unke beech ke angle par depend karta hai. [Q1253, Q1256, Q1292]
- Work done initial velocity par depend nahi karta. [Q1292]
- Power = Work / Time; kaun kitni jaldi kaam complete karta hai ye batata hai. [Q1282]

7. Scalar vs Vector Quantities

- Scalar (sirf magnitude): Work, power, pressure, energy, frequency, speed, mass, temperature. [Q1288]
- Vector (magnitude + direction): Displacement, force, torque, momentum, acceleration, velocity. [Q1288]

8. Temperature, States of Matter & Particle Energy

- Temperature badhane par solid/liquid/gas ke particles ki kinetic energy increases. [Q1225]
- Kelvin temperature directly proportional to average kinetic energy of particles. [Q1225]
- Kisi bhi substance ke particles same temperature par different amounts of KE rakhte hain (Maxwell-Boltzmann distribution). [Q1228]
- Gases mein particles ki KE maximum hoti hai; solids mein minimum. [Q1290]

9. Conservation of Energy & Miscellaneous Facts

- Law of Conservation of Energy: Energy ko na toh create kar sakte hain na destroy — sirf ek form se doosre mein convert kar sakte hain. [Q1264, Q1269, Q1300]
- Explosion mein fragments alag direction mein jaate hain lekin momentum conserved rehta hai; KE conserved nahi hoti. [Q1233]
- Power scalar quantity hai. [Q1221]
- Telephone ka invention 1876 mein Alexander Graham Bell ne kiya. [Q1218]
- Ultrasound: >20 kHz frequency; defects se reflect back hota hai (bats use karte hain). [Q1220]

10. ⚠ Traps / Formula Cheat Sheet (MCQ Q1210–Q1314)

- ⚠ Trap: $P = IR^2$ galat formula hai — correct hai $P = I^2R$. [Q1210]
- ⚠ Trap: India mein 220V, 50 Hz hai — 110V/60Hz USA ka hai. [Q1213]
- ⚠ Trap: Work done mass par depend nahi karta — ye common confusion hai. [Q1253, Q1256, Q1292]
- ⚠ Trap: Free fall mein ME constant rehti hai lekin KE badhti, PE ghatti hai; dono opposite direction mein change hoti hain. [Q1237, Q1244, Q1298]
- ⚠ Trap: Same momentum par bullet ki KE gun se zyada hoti hai kyunki $KE = p^2/2m$ — kam mass = zyada KE. [Q1242, Q1266]
- ⚠ Trap: Raised hammer mein PE hai, KE nahi — motion nahi hai. [Q1254, Q1270, Q1312]
- ⚠ Trap: Wind opposing car ka work negative hota hai, positive nahi. [Q1251]
- ⚠ Trap: $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$ — power nahi, energy hai. [Q1246, Q1252]
- ⚠ Trap: Circular motion mein aur Moon ki parikrama mein work zero hota hai kyunki force hamesha perpendicular hoti hai displacement se. [Q1296, Q1303]
- ⚠ Trap: Running water KE hai, PE nahi — stored dam water PE hai. [Q1285, Q1312]
- ⚠ Trap: Friction ka work hamesha negative — kabhi positive nahi. [Q1289]
- ⚠ Trap: Car ki velocity badhne par uski PE same rehti hai agar height constant ho; confuse mat hona. [Q1293]
- ⚠ Trap: Bicycle chalate waqt chemical energy nahi hoti directly; heat, KE, aur mechanical energy hoti hai. [Q1295]
- ⚠ Trap: $1 \text{ HP} = 746 \text{ W}$ yaad rakhna — 1000 W nahi. [Q1311]
- ⚠ Trap: Hydroelectric mein ultimately $KE \rightarrow \text{Electrical}$ hota hai, $PE \rightarrow \text{Electrical}$ directly nahi (PE pehle KE mein convert hoti hai flow ke saath). [Q1307]
- ⚠ Trap: 200 m se 100 m girne par "more KE less PE" nahi balki donon barabar (half-half) hote hain total ME constant hone ki wajah se. [Q1280]
- ⚠ Trap: Natural gas conventional source hai — solar/wind/nuclear ko isse alag samajhna. [Q1281]
- ⚠ Trap: Bulb ki 100 W rating ka matlab 100 Joule per second hai — per minute ya per hour nahi. [Q1226]

- ⚠ Trap: 1 joule/second = 1 watt — energy ka unit joule, power ka unit watt confuse mat karna. [Q1268]
- ⚠ Trap: Household/commercial energy consumption kWh mein measure hoti hai, watt ya joule mein nahi. [Q1272]

11. Numerical: Work Done by Force

- Basic formula: Work (W) = Force (F) × Displacement (d) jab dono same direction mein ho. [Q1327, Q1340, Q1351, Q1358, Q1363, Q1391, Q1396, Q1406, Q1417]
- 10 N force se 2 m displacement → W = 20 J. [Q1327]
- 50 N force se 10 m displacement → W = 500 J. [Q1340]
- 10 N force se 5 m displacement → W = 50 J. [Q1351]
- 20 N force se 4 m displacement → W = 80 J. [Q1358]
- 125 N force se 5 m displacement → W = 625 J. [Q1363]
- 25 N force se 5 m displacement → W = 125 J. [Q1391]
- 10 N force se 10 m displacement → W = 100 J. [Q1396]
- 10 N force se 5 m displacement → W = 50 J. [Q1406]
- 1 N force se 1 m displacement → W = 1 Joule (SI unit definition). [Q1417]
- 120 N box ko 2 m upar uthana → W = 240 J. [Q1422]
- 50 N force se 100 m displacement → W = 5000 J. [Q1368]
- Plane 4000 m distance with 20000 J work → Force = W/d = 5 N. [Q1402]
- Object vertically u velocity se floor se takraye aur same velocity se bounce back → change in motion = 0. [Q1389]

12. Numerical: Kinetic Energy — Formula & Variations

- Formula: KE = $\frac{1}{2}mv^2$; unit Joule (J). [Q1315, Q1323, Q1328, Q1330, Q1334, Q1336, Q1341, Q1343, Q1344, Q1346, Q1347, Q1354, Q1360, Q1362, Q1364, Q1365, Q1366, Q1367, Q1369, Q1370, Q1371, Q1407, Q1408, Q1410, Q1411, Q1412, Q1414, Q1418, Q1420, Q1425]
- 1 kg object 4 m/s se 8 m/s → Work = $\Delta KE = \frac{1}{2} \times 1 \times (64 - 16) = 24$ J. [Q1315]
- 800 kg car 5 m/s se 10 m/s → W = $\frac{1}{2} \times 800 \times (100 - 25) = 30,000$ J = 30 kJ. [Q1328]
- 10 kg block rest se 2 m/s in 20 s → Work = $\frac{1}{2} \times 10 \times 4 = 20$ J; Power = 20/20 = 1 W. [Q1326]
- 20 kg object 8 m/s → KE = $\frac{1}{2} \times 20 \times 64 = 640$ J. [Q1341]
- 10 kg object 6 m/s → KE = $\frac{1}{2} \times 10 \times 36 = 180$ J. [Q1343]
- Object 4 m/s par KE = 120 J → mass = $2 \times 120 / 16 = 15$ kg. [Q1344, Q1347]
- Object 5 m/s par KE = 25 J; velocity double → KE = 100 J. [Q1346]
- 2 kg ball 30 m/s → KE = $\frac{1}{2} \times 2 \times 900 = 900$ J. [Q1354]
- 20 kg object 10 m/s → KE = $\frac{1}{2} \times 20 \times 100 = 1000$ J. [Q1360]
- 15 kg object 6 m/s → KE = $\frac{1}{2} \times 15 \times 36 = 270$ J. [Q1362]
- 15 kg object 8 m/s → KE = $\frac{1}{2} \times 15 \times 64 = 480$ J. [Q1364]
- KE = 120 J, mass = 15 kg → velocity = $\sqrt{(2 \times 120 / 15)} = 4$ m/s. [Q1365]
- 15 kg object 2 m/s → KE = $\frac{1}{2} \times 15 \times 4 = 30$ J. [Q1366]
- 10 kg object 5 m/s → KE = $\frac{1}{2} \times 10 \times 25 = 125$ J. [Q1367]
- 13 kg object 5 m/s → KE = $\frac{1}{2} \times 13 \times 25 = 162.5$ J. [Q1369]
- 14 kg object 5 m/s → KE = $\frac{1}{2} \times 14 \times 25 = 175$ J. [Q1370]

- 30 kg object 5 m/s $\rightarrow KE = \frac{1}{2} \times 30 \times 25 = 375 \text{ J}$. [Q1371]
- 10 kg object 6 m/s $\rightarrow KE = \frac{1}{2} \times 10 \times 36 = 180 \text{ J}$. [Q1407]
- 4 kg object 5 m/s se 10 m/s $\rightarrow Work = \frac{1}{2} \times 4 \times (100 - 25) = 150 \text{ J}$. [Q1408]
- 4 kg object rest se 5 m/s $\rightarrow Work = \frac{1}{2} \times 4 \times 25 = 50 \text{ J}$. [Q1410]
- 0.1 kg ball 2 m girne par gravity ka work = mgh = $0.1 \times 9.8 \times 2 = 1.96 \text{ J}$. [Q1412]
- Same velocity par mass double $\rightarrow KE$ double hoti hai (KA/KB = 2). [Q1323]
- Velocity double karne par KE 4 times ho jaati hai. [Q1330, Q1336, Q1346, Q1411, Q1418]
- 1 kg object ka KE = 1 J ho to velocity = $\sqrt{2} \approx 1.4 \text{ m/s}$. [Q1401]
- 5 kg aur 10 kg balls ka KE barabar ho $\rightarrow 5 \text{ kg ball tez chalti hai kyunki } v^2 = 2KE/m$. [Q1420]
- KE 256 times ho jaye to momentum 16 times ho jaata hai (kyunki $KE \propto p^2$). [Q1414]
- Momentum triple ho to KE 9 times ho jaati hai. [Q1425]
- Momentum double ho to KE 4 times ho jaati hai. [Q1334]

13. Numerical: Potential Energy — Formula & Height Problems

- Formula: $PE = mgh$; unit Joule (J). [Q1316, Q1318, Q1320, Q1325, Q1331, Q1333, Q1335, Q1337, Q1339, Q1342, Q1348, Q1349, Q1350, Q1352, Q1353, Q1355, Q1356, Q1357, Q1359, Q1361, Q1372, Q1373, Q1374, Q1375, Q1376, Q1378, Q1379, Q1381, Q1384, Q1385, Q1395, Q1397, Q1398, Q1399, Q1405, Q1415, Q1416, Q1423]
- 20 kg luggage 2 m upar $\rightarrow W = mgh = 20 \times 10 \times 2 = 400 \text{ J}$. [Q1316, Q1335]
- 9800 J energy se 70 kg mass ko height = $E/(mg) = 9800/(70 \times 10) = 14 \text{ m}$. [Q1318]
- 6 kg object PE = 480 J $\rightarrow$ height = $480/(6 \times 10) = 8 \text{ m}$. [Q1320]
- 1 kg object 10 m upar uthana $\rightarrow$ gravity ka work = $-mgh = -98 \text{ J}$ (negative kyunki gravity opposite direction mein kaam karta hai). [Q1325]
- 50 kg object 6 m height par ($g=10$) $\rightarrow PE = 50 \times 10 \times 6 = 3000 \text{ J}$. [Q1331]
- 1 kg object PE = 2 J $\rightarrow$ height = $2/(1 \times 10) = 0.2 \text{ m}$. [Q1333]
- 0.5 kg mobile 100 m height $\rightarrow PE = 0.5 \times 10 \times 100 = 500 \text{ J}$. [Q1337]
- 10 kg object 6 m height ($g=9.8$) $\rightarrow PE = 10 \times 9.8 \times 6 = 588 \text{ J}$. [Q1339, Q1395]
- 20 kg object PE = 400 J ($g=10$) $\rightarrow$ height = $400/(20 \times 10) = 2 \text{ m}$. [Q1342]
- 12 kg object PE = 480 J ($g=10$) $\rightarrow$ height = $480/120 = 4 \text{ m}$. [Q1348, Q1384]
- 11 kg luggage 1.5 m upar $\rightarrow W = 11 \times 10 \times 1.5 = 165 \text{ J}$. [Q1349]
- 12 kg luggage 1.5 m upar $\rightarrow W = 12 \times 10 \times 1.5 = 180 \text{ J}$. [Q1350]
- 12 kg object PE = 600 J $\rightarrow$ height = $600/120 = 5 \text{ m}$. [Q1352, Q1379]
- 14 kg luggage 1.5 m upar $\rightarrow W = 14 \times 10 \times 1.5 = 210 \text{ J}$. [Q1353]
- 15 kg hammer 10 m upar $\rightarrow PE = 15 \times 10 \times 10 = 1500 \text{ J}$. [Q1355]
- 13 kg luggage 1.5 m upar $\rightarrow W = 13 \times 10 \times 1.5 = 195 \text{ J}$. [Q1356]
- 15 kg luggage 1.0 m upar $\rightarrow W = 15 \times 10 \times 1 = 150 \text{ J}$. [Q1357]
- 10 kg luggage 1.1 m upar $\rightarrow W = 10 \times 10 \times 1.1 = 110 \text{ J}$. [Q1359]
- 10 kg luggage 1.2 m upar $\rightarrow W = 10 \times 10 \times 1.2 = 120 \text{ J}$. [Q1361]
- 12 kg object 5 m height ($g=9.8$) $\rightarrow PE = 12 \times 9.8 \times 5 = 588 \text{ J}$. [Q1372]
- 15 kg object 5 m height ($g=9.8$) $\rightarrow PE = 15 \times 9.8 \times 5 = 735 \text{ J}$. [Q1373]
- 11 kg object 6 m height ($g=9.8$) $\rightarrow PE = 11 \times 9.8 \times 6 = 646.8 \text{ J}$. [Q1374]
- 11 kg object 8 m height ($g=9.8$) $\rightarrow PE = 11 \times 9.8 \times 8 = 862.4 \text{ J}$. [Q1375]
- Mass m ko 4h height par $\rightarrow PE = 4mgh$. [Q1376]
- 10 kg object 7 m height ($g=9.8$) $\rightarrow PE = 10 \times 9.8 \times 7 = 686 \text{ J}$. [Q1378]

- 11 kg object 7 m height ($g=9.8$) $\rightarrow PE = 11 \times 9.8 \times 7 = 754.6 \text{ J}$. [Q1381]
- 10 kg object 4 m girne par gravity ka work = $mgh = 10 \times 10 \times 4 = 400 \text{ J}$. [Q1397]
- Object 5 m height par $PE = 2000 \text{ J} \rightarrow \text{mass} = 2000 / (10 \times 5) = 40 \text{ kg}$. [Q1398]
- 3m mass ko h height par $\rightarrow PE = 3mgh$. [Q1399]
- 20 kg object 5 m upar $\rightarrow \text{gravity ka work} = mgh = 1000 \text{ J}$. [Q1405]
- 20 kg object 2 m upar $\rightarrow \text{work} = 20 \times 10 \times 2 = 400 \text{ J}$. [Q1415]
- 40 kg Mohan 50 steps (15 cm each) in 10 s $\rightarrow PE = 40 \times 10 \times 7.5 = 3000 \text{ J}$. [Q1416]

14. Numerical: Power — Human Climbing & Engine Problems

- Formula: Power (P) = Work/Time = mgh/t ; unit Watt (W). [Q1317, Q1319, Q1321, Q1322, Q1326, Q1329, Q1332, Q1377, Q1385, Q1386, Q1387, Q1388, Q1393, Q1416, Q1421]
- 1000 J work in 2 s $\rightarrow \text{Power} = 1000/2 = 500 \text{ W}$. [Q1317]
- 40 kg girl 5 m in 4 s $\rightarrow \text{Power} = mgh/t = 40 \times 10 \times 5/4 = 500 \text{ W}$. [Q1319]
- 80 kg object 40 m in 50 s $\rightarrow \text{Power} = 80 \times 10 \times 40/50 = 640 \text{ J/s (W)}$. [Q1321]
- 400 N girl 160 W power se 20 s tak climb $\rightarrow \text{height} = (160 \times 20)/400 = 8 \text{ m}$. [Q1322]
- 10 kg block rest se 2 m/s in 20 s $\rightarrow \text{Work} = 20 \text{ J}$; Power = 1 W. [Q1326]
- 20 kW engine 10 s $\rightarrow \text{Work} = 20 \times 10 = 200 \text{ kJ}$. [Q1329]
- 50 kg boy 45 stairs (15 cm) in 9 s $\rightarrow \text{Power} = 50 \times 10 \times 6.75/9 = 375 \text{ W}$. [Q1332, Q1377]
- 50 kg boy 40 stairs (15 cm) in 10 s $\rightarrow \text{Power} = 50 \times 10 \times 6/10 = 300 \text{ W}$. [Q1385]
- 50 kg boy 45 stairs (16 cm) in 10 s $\rightarrow \text{Power} = 50 \times 10 \times 7.2/10 = 360 \text{ W}$. [Q1386]
- 50 kg boy 40 stairs (15 cm) in 9 s $\rightarrow \text{Power} = 50 \times 10 \times 6/9 = 333.33 \text{ W}$. [Q1387]
- 50 kg boy 43 stairs (15 cm) in 10 s $\rightarrow \text{Power} = 50 \times 10 \times 6.45/10 = 322.5 \text{ W}$. [Q1388]
- 50 kg boy 41 stairs (15 cm) in 10 s $\rightarrow \text{Power} = 50 \times 10 \times 6.15/10 = 307.5 \text{ W}$. [Q1393]
- 40 kg Mohan 50 stairs (15 cm) in 10 s $\rightarrow \text{Power} = 40 \times 10 \times 7.5/10 = 300 \text{ W}$. [Q1416]
- Boy 20 m in 4 min (240 s) with 600 N $\rightarrow \text{Power} = (600 \times 20)/240 = 50 \text{ W}$. [Q1421]
- If a factor W does work in time t, then its power $P = \text{Work/Time (W/t)}$. [Q1345]

15. Numerical: Work-Energy Theorem

- Work done = Change in KE = $\frac{1}{2}m(v^2 - u^2)$. [Q1315, Q1328, Q1408, Q1410]
- 1 kg object 4 $\rightarrow$ 8 m/s $\rightarrow W = \frac{1}{2} \times 1 \times (64 - 16) = 24 \text{ J}$. [Q1315]
- 800 kg car 5 $\rightarrow$ 10 m/s $\rightarrow W = \frac{1}{2} \times 800 \times 75 = 30,000 \text{ J} = 30 \text{ kJ}$. [Q1328]
- 4 kg object 5 $\rightarrow$ 10 m/s $\rightarrow W = \frac{1}{2} \times 4 \times 75 = 150 \text{ J}$. [Q1408]
- 4 kg object rest $\rightarrow$ 5 m/s $\rightarrow W = \frac{1}{2} \times 4 \times 25 = 50 \text{ J}$. [Q1410]
- 2 kg object upward 20 m/s, after 2 s ($g=10$): $v = 20 - 20 = 0 \rightarrow KE = 0 \text{ J}$. [Q1409]
- Ball 10 m se girne ke baad 2.5 m tak bounce back $\rightarrow KE \text{ loss} = (10 - 2.5)/10 \times 100 = 75\%$. [Q1413]

16. Numerical: Electrical Work — $W = VQ$

- Formula: Work (W) = Potential Difference (V) $\times$ Charge (Q). [Q1380, Q1382, Q1383, Q1390, Q1392, Q1394]
- $V = 20 \text{ V}$, $W = 60 \text{ J} \rightarrow Q = W/V = 60/20 = 3 \text{ C}$. [Q1380]

- $Q = 4 \text{ C}, V = 10 \text{ V} \rightarrow W = 10 \times 4 = 40 \text{ J}$. [Q1382]
- $Q = 3 \text{ C}, W = 72 \text{ J} \rightarrow V = 72/3 = 24 \text{ V}$. [Q1383]
- $W = 20 \text{ J}, V = 10 \text{ V} \rightarrow Q = 20/10 = 2 \text{ C}$. [Q1390]
- $Q = 4 \text{ C}, V = 12 \text{ V} \rightarrow W = 12 \times 4 = 48 \text{ J}$. [Q1392]
- $Q = 5 \text{ C}, V = 13 \text{ V} \rightarrow W = 13 \times 5 = 65 \text{ J}$. [Q1394]

17. Numerical: Elastic Potential Energy

- Formula: Elastic Energy = $\frac{1}{2} \times F \times x$ (x = extension). [Q1404]
- $F = 200 \text{ N}, x = 1 \text{ mm} = 10^{-3} \text{ m} \rightarrow \text{Energy} = \frac{1}{2} \times 200 \times 10^{-3} = 0.1 \text{ J}$. [Q1404]

18. Numerical: Inclined Plane Work

- Smooth inclined plane par work: $W = mg \sin \theta \times d$. [Q1400]
- 2 kN block, 10 m displacement, 15° angle $\rightarrow W = 2000 \times \sin 15^\circ \times 10 = 2000 \times 0.26 \times 10 = 5.17 \text{ kJ}$. [Q1400]

19. Numerical: Momentum-KE Relationship

- Formula: $KE = p^2/(2m)$ jahan p = momentum. [Q1403]
- Momentum p wale ball ki $KE = p^2/2m$. [Q1403]
- Momentum double $\rightarrow KE$ 4 times. [Q1334]
- Momentum triple $\rightarrow KE$ 9 times. [Q1425]
- KE 256 times $\rightarrow$ Momentum 16 times ($\sqrt{256} = 16$). [Q1414]

20. Numerical: Efficiency & Energy Conversion

- Efficiency: $\eta = (\text{Output}/\text{Input}) \times 100\%$. [Q1324]
- Input = 2000 J, Output = 500 J $\rightarrow \eta = (500/2000) \times 100 = 25\%$. [Q1324]
- 320 units energy = 320 kWh = $320 \times 3.6 \times 10^6 = 1152 \times 10^6 \text{ J}$. [Q1424]

21. Numerical: Formula Cheat Sheet

- Work (Force parallel): $W = F \times d$ — unit Joule (J).
- Work (against gravity): $W = mgh$ — unit Joule (J).
- Kinetic Energy: $KE = \frac{1}{2}mv^2 = p^2/2m$ — unit Joule (J).
- Potential Energy: $PE = mgh$ — unit Joule (J).
- Power: $P = W/t = mgh/t$ — unit Watt (W).
- Work-Energy Theorem: $W = \Delta KE = \frac{1}{2}m(v^2 - u^2)$ — unit Joule (J).
- Electrical Work: $W = VQ = VIt$ — unit Joule (J).
- Elastic Energy: $E = \frac{1}{2}Fx$ — unit Joule (J).

- Efficiency: $\eta = \text{Output/Input} \times 100\%$ — unit %.
- 1 kWh = 3.6×10^6 J.

22. ⚠ Traps / Key Differences Summary (Numerical Q1315–Q1425)

- ⚠ Trap: Object ko upar uthane par gravity ka work negative ($-mgh$) hota hai; neeche girne par positive ($+mgh$). [Q1325, Q1397]
- ⚠ Trap: Velocity double karne par KE 4 times hoti hai, 2 times nahi. [Q1330, Q1336, Q1346, Q1411, Q1418]
- ⚠ Trap: Momentum double karne par KE 4 times hoti hai (kyunki $KE \propto p^2$). [Q1334]
- ⚠ Trap: Momentum triple karne par KE 9 times hoti hai. [Q1425]
- ⚠ Trap: KE 256 times ho jaye to momentum 16 times hota hai ($\sqrt{256} = 16$), 256 times nahi. [Q1414]
- ⚠ Trap: Equal KE wale objects mein kam mass wala tez chalta hai. [Q1420, Q1323]
- ⚠ Trap: Work zero jab displacement zero ho, chahe kitna bhi force lage. [Q1338, Q1419]
- ⚠ Trap: 2 kg object 20 m/s se upar, 2 s baad $v = 0 \rightarrow KE = 0$ J (highest point par). [Q1409]
- ⚠ Trap: Steps ka height cm mein hota hai — hamesha meter mein convert karo (divide by 100). [Q1332, Q1377, Q1385, Q1386, Q1387, Q1388, Q1393, Q1416]
- ⚠ Trap: Time minutes mein de sakta hai — seconds mein convert karo ($\times 60$). [Q1421]
- ⚠ Trap: 320 units = 320 kWh = 1152×10^6 J; $320 \times 3.6 \times 10^6$ mat bhoolna. [Q1424]
- ⚠ Trap: Object vertically u velocity se bounce back kare to change in motion = 0 (same speed magnitude). [Q1389]
- ⚠ Trap: Ball 10 m se girke 2.5 m bounce back kare to 75% KE loss hoti hai $((10-2.5)/10)$. [Q1413]
- ⚠ Trap: 1 Joule = $1 \text{ N} \times 1 \text{ m}$ — ye definition yaad rakhna. [Q1417]
- ⚠ Trap: Work ka unit hamesha Joule hota hai — options mein W, N, Pa, C confuse karne ke liye hote hain. [Q1341, Q1354, Q1360, Q1363, Q1405, Q1406]
- ⚠ Trap: Power ka formula "time-work" nahi balki Work/Time hota hai — order confuse mat karna. [Q1345]